

SPHERE CARE

Video and AI Aged Care Platform

Meeting Agendas and Minutes

CSIT321 Software Engineering Capstone | University of Wollongong
Academic Year 2025 – 2026

Project	Sphere Care — AI-Powered Aged Care Platform													
Document	Meeting Agenda													
Version	1.0 (Created at 26/5/2026)													
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Meetings Overview

This document contains the agenda and recorded minutes for all group meetings held during the Sphere Care CSIT321 capstone project. Meetings were conducted via Zoom (online) or on campus depending on availability. No formal meetings were held during the mid-semester break.

Team Communication Channels: Discord (internal team) , Whatsapp (communication with Supervisor)

No.	Type / Format	Date	Location	Key Topic
M1	Online — Zoom	Week 1, Sem 1	Zoom	Project introduction and scope
M2	Online — Zoom	Week 3, Sem 1	Zoom	Task allocation and role assignment
M3	On Campus	Week 5-7, Week 9 Sem 1	Building 6	Figma design, user stories, storyboard
M4	Online — Zoom	Week 8, Sem 1	Zoom (QA)	QA review with supervisor and other groups
M5	Online — Discord	Week 1, Sem 2	Zoom	Frontend development kickoff (web and mobile)
M6	On Campus	Week 3, Sem 2	Building 6	Draft UI review with supervisor
M7	On Campus	Week 5-9, Sem 2	Library	ASL motion recording session
M8	Online — Zoom	Week 10-11, Sem 2	Zoom	Progress check and integration review
M9	Online — Zoom	Week 13, Sem 2	Zoom	Final sprint review and submission prep

Note: Several meetings to discuss video/demo content were proposed but could not proceed due to supervisor unavailability. These are noted in the relevant sections below.

Meeting 1

Date: Week 1, Semester 1 Venue: Zoom Attendees: All team members + Supervisor

Agenda

- Welcome and team introductions
- Overview of the CSIT321 capstone project structure and requirements
- Introduction to the Sphere Care project brief from the supervisor
- Discussion of project goals, expected deliverables, and timeline
- Initial questions and clarifications

Minutes

Chairperson: Aneta Guzowska(Supervisor)

Scribe: Yu Han

- Supervisor introduced the project — Sphere Care is an AI-assisted care communication and documentation platform for aged care facilities.
- The team was briefed on the project scope: the system would involve video recording, AI transcription, safety detection, messaging, and family communication features.
- Deliverables were outlined: a staff web application, a client mobile application, a FastAPI backend, and complete project documentation.
- The supervisor explained the marking criteria including requirements documentation, sprint management, testing, and final demonstration.
- Team members introduced themselves and their general technical backgrounds.
- The next meeting was scheduled for the following week to discuss task allocation.

Action Items

1. All members: join the team Discord server and confirm access.
2. Aneta: share additional project documents to clarify platform requirements.

Meeting 2

Date: Week 2, Semester 1 Venue: Zoom Attendees: All team members + Supervisor

Agenda

- Review of project requirements and scope from Meeting 1
- Discuss and agree on team roles and responsibilities
- Allocate features and components to individual members
- Agree on development tools, framework choices, and coding conventions
- Set timeline for Sprint 1 deliverables

Minutes

Chairperson: Aneta Guzowska (Supervisor)

Scribe: Yu Han

- The team reviewed the project scope and confirmed core features to be built: authentication, resident management, recording, transcription, AI flagging, messaging, calls, and bookings.
- Roles and responsibilities were allocated as follows:
 - Araf Akhter — Database, security, configuration
 - ChenCheng Lu — Backend core, LiveKit WebRTC
 - Chingyu Chen — Frontend client mobile app
 - Lite Ye — Backend Core, Whisper
 - Tonmoy Saker — AI vision pipeline, YOLO/MediaPipe, safety flagging
 - Yu Han Yong — Frontend staff web
- Technology stack was agreed: FastAPI (Python), PostgreSQL, React Native/Expo (mobile), Bootstrap 5 (staff web), LiveKit WebRTC, Whisper ASR, YOLOv8.
- Git branching strategy was agreed: feature branches merged via pull requests to main.
- Sprint 1 goal: set up development environments, confirm requirements, create initial wireframes.

Action Items

3. All members: set up a local development environment and confirm connection to a shared GitHub repository.
4. All members: document initial requirements for their allocated features.

Meeting 3

Date: Week 3, Semester 1 Venue: On Campus Attendees: All team members + Supervisor

Agenda

- Walk through user stories for each user role (Admin, Staff, Client/Family)
- Present and discuss storyboard for key user journeys
- Review QA process and testing expectations
- Confirm Sprint 2 goals

Minutes

Chairperson: Aneta Guzowska (Supervisor)

Scribe: Yu Han Yong

- User stories were walked through for each role:
 - Admin: manage users, view audit logs, configure centre settings, export compliance reports.
 - Staff: view residents, record care sessions, review AI flags, message families, manage bookings.
 - Client/Family: receive messages, join video/audio calls, view approved care updates, manage bookings.
- Storyboards were presented for two key journeys: a staff member recording a care session and a family member receiving a safety alert notification.
- The supervisor clarified that AI outputs must be treated as support tools requiring staff review — no fully automated decisions.

Action Items

5. All members start to do figma wireframe for both mobile and website.

Meeting 4

Date: Week 4, Semester 1 Venue: Zoom (QA Session) Attendees: All team members + Supervisor + Other CSIT321 Groups

Agenda

- QA review session with supervisor and other CSIT321 project groups

Minutes

Chairperson: Aneta Guzowska (Supervisor)

Format: Joint QA session with multiple CSIT321 project groups

- This meeting was a joint QA session organised by the supervisor, attended by multiple CSIT321 project groups.
- The Sphere Care team presented their project overview, scope and user roles
- Key questions from peer groups included:
 - How does the AI safety detection pipeline work with live video vs recorded clips?
 - How is privacy handled for recording of aged care residents?
- The supervisor used the session to clarify assessment expectations for all groups: requirements documentation, sprint evidence, testing records, and a functional demonstration are all required for final submission.
- Feedback from the session was noted for incorporation into the requirements document.

Action Items

6. All members: review QA session notes and update requirements documentation where needed.

Mid-Semester Break — No Meetings Held

Meeting 5

Date: Week 1, Semester 2 Venue: Zoom Attendees: All team members

Agenda

- Resume project after mid-semester break
- Review Sprint 2 deliverables completed before break
- Plan Sprint 3: frontend development kickoff for staff web and mobile client
- Discuss API integration plan between frontend and backend
- Confirm timeline for Semester 2

Minutes

Chairperson: Yu Han Yong

Scribe: Tonmoy Sarker

- The team resumed the project after the mid-semester break. The supervisor acknowledged the break and confirmed the updated timeline for Semester 2 deliverables.
- Sprint 2 review: backend foundations (auth, residents, records, flags endpoints) were completed. Database schema was finalised.
- Sprint 3 goals were agreed:
 - Yu Han Yong: begin building HTML/CSS/JS pages for the staff web app based on finalised Figma designs.
 - Chingyu Chen: begin Expo/React Native setup for the client mobile app.
 - All backend members: begin frontend-to-backend API integration and ensure all endpoints return correct response schemas.
- API integration approach was confirmed: the staff web app and mobile client will connect to the same FastAPI backend. All endpoints require JWT Bearer token authentication.
- The WebSocket layer for real-time messaging and notifications was identified as a high-priority item for Sprint 3.

Action Items

7. All members: ensure API responses match the agreed frontend data contracts.

Meeting 6

Date: Week 3, Semester 2 Venue: University of Wollongong Campus Attendees: All team members + Supervisor

Agenda

- Live demo of current draft frontend pages to supervisor
- Walkthrough of staff web application pages built so far
- Walkthrough of client mobile application screens
- Supervisor feedback on UI, layout, and feature completeness
- Identify gaps and prioritise remaining work for Sprint 4

Minutes

Chairperson: Aneta Guzowska (Supervisor)

Scribe: Tonmoy Sarker

Format: On campus

- The team demonstrated current draft pages from the staff web application to the supervisor in person, including the dashboard, residents list, records library, flags board, and recording console.
- The client mobile application was shown running on Expo web with home, messages, and booking screens.
- Supervisor feedback on the staff web application:
 - Dashboard metrics were clear and well-organised.
 - The flags board Kanban layout was approved — severity colour coding was positively noted.
 - Requested that the recording console clearly communicate recording status to avoid confusion for care staff.
 - Suggested adding a search bar to the residents list for larger facilities.
- Supervisor feedback on the mobile client:
 - Navigation was considered appropriate for the target user group.
 - Requested that call invite screens be clearly distinguishable from regular notifications.

Action Items

8. All members: continue Sprint 4 development based on supervisor feedback.

Meeting 7

Date: Week 5- Week 9, Semester 2 Venue: University of Wollongong Campus Attendees: All team members

Agenda

- On-campus ASL motion recording session for AI training and testing data
- Set up recording equipment and test the recording console
- Record ASL gesture sequences for use in model evaluation
- Review recording quality and discuss any technical issues
- Plan next steps for AI pipeline testing with recorded data

Minutes

Chairperson: Tonmoy Sarker

Format: Practical on-campus recording session

- The team conducted an on-campus recording session to capture ASL (American Sign Language) motion data for use in testing the AI vision pipeline.
- The recording console was set up using the staff web application with camera feed connected.
- Team members took turns performing gesture sequences in front of the camera. Sequences included hand movements, gestures, and basic motion patterns relevant to the fall and safety detection scenarios.
- Tonmoy Saker and Araf Arkhter supervised the recording setup and confirmed clips were being saved correctly to the backend.
- Several clips were processed through the Whisper ASR pipeline to test audio quality detection alongside motion capture.
- Minor issues identified:
 - Camera frame rate dropped below target when running in some browser environments — noted for optimisation.
 - Recording status indicator was not updating reliably in one scenario — flagged for fix.
- Recorded clips were retained for use in AI pipeline integration testing in Sprint 5.

Action Items

9. All members: review Sprint 5 goals at the next meeting.

Meeting 8

Date: Week 10- Week 11, Semester 2 Venue: Zoom Attendees: All team members

Agenda

- Sprint 5 progress update from all team members
- Review integration between frontend and backend
- Discuss AI pipeline testing results from ASL recording session
- Address any remaining bugs and technical blockers
- Discuss video/demo content production (tentative)

Minutes

Chairperson: Tonmoy Sarker

Scribe: Yu Han

- Sprint 5 progress was reviewed. All core features were confirmed as implemented and connected end-to-end.
- Integration status confirmed:
 - Staff web app: all pages connected to backend API including messaging, bookings, records, and flags.
 - Mobile client: login, messaging, notifications, calls, and bookings connected to backend.
 - WebSocket real-time delivery confirmed working for messages and notifications.
- Tonmoy reported that the YOLO/MediaPipe pipeline successfully detected motion events from the ASL recording session clips. Results were consistent with expected outputs.
- Known issues discussed:
 - Mobile push notifications require production Expo credentials — confirmed as a known limitation for prototype scope.
 - LiveKit production setup deferred — call workflow and signalling is functional without full media connection.
- The team raised the topic of video/demo content for the final presentation and TechExpo. The supervisor indicated availability was limited and would try to confirm a time for a separate session.
- Sprint 6 goals agreed: final bug fixes, documentation completion, testing records, and submission preparation.

Action Items

10. All members: complete Sprint 6 documentation and testing evidence.
11. All members: follow up on video/demo scheduling — supervisor to confirm availability.

Meeting 9

Date: Week 13, Semester 2 Venue: Zoom Attendees: All team members + Supervisor

Agenda

- Final sprint review — confirm all deliverables are complete
- Review documentation package (technical report, user manual, source code, minutes)
- Confirm submission checklist
- Discuss TechExpo demonstration plan
- Final supervisor sign-off

Minutes

Chairperson: Aneta Guzowska (Supervisor)

Scribe: Yu Han Yong

- The team presented a final status update confirming all Core and Additional requirements had been implemented and tested.
- Documentation review:
 - Technical report: complete including architecture, testing, installation, and data dictionaries.
 - User manual: complete with screenshots for all staff web and mobile client workflows.
 - Source code documentation: complete with professional comments and author attribution.
 - Meeting minutes: this document, covering all 9 recorded meetings.
 - Individual reflections: in progress, to be submitted separately.
- The submission checklist was reviewed and all items confirmed as ready or in final preparation.
- TechExpo demonstration plan was discussed:
 - The team will demonstrate the full end-to-end workflow: staff login, resident profile, recording, ASR transcription, AI flag generation, flag review, family messaging, and booking management.
 - The client mobile app will also be shown on a separate device.
- The supervisor confirmed satisfaction with the project outcome and indicated that the final product met the project objectives.
- The video/demo content sessions that were requested earlier could not proceed due to supervisor scheduling constraints. The team noted this in the project closeout documentation.
- The supervisor thanked the team for their effort throughout the project and wished them well for submission.

Action Items

12. All members: prepare TechExpo demonstration setup.

Note: Requested Meetings Not Held

The following meeting attempts were made but could not proceed due to supervisor unavailability.

During Semester 2, the team identified the need for dedicated sessions to discuss video content production and the final demonstration video. The following meeting requests were made but could not be confirmed:

Request	Purpose	Requested By	Outcome
Request 1	Review and approve demo video script before recording	Full team	Supervisor unavailable — not held
Request 2	Video walkthrough session for final submission demo content	Full team	Supervisor unavailable — not held

The team proceeded with video content preparation independently and incorporated available supervisor feedback from other meetings into the demonstration content.

— End of Meeting Agendas and Minutes —

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